

# Martim Ramos

**Mobile:** +351 933 694 114

**Email:** [m@martimcr.com](mailto:m@martimcr.com)

**LinkedIn:** <https://www.linkedin.com/in/martim-c-ramos>

**Location:** Lisbon, Portugal

**Portfolio & Code:** <https://github.com/ramos-99> | <https://martimcr.com>

## SUMMARY

Computer engineer (IST Lisbon), starting a Master's in AI and Infrastructures in September 2026. Builds low-level systems software in Rust and eBPF, including truetop, a TUI system monitor, and contributions to the aya-rs eBPF framework. Co-founder of RQ Systems.

## EDUCATION

**Bachelor of Science, Computer Science and Engineering (LEIC)** - Instituto Superior Técnico, Lisbon | Sep 2023 – Jul 2026 | Final grade: 14.5/20

**Master's, AI and Infrastructures (MEIC)** - Instituto Superior Técnico, Lisbon | Starting Sep 2026

## OPEN SOURCE CONTRIBUTIONS & PROJECTS

**truetop - Personal Project** | Jun 2026 – Present

- Built a per-process system monitor (Rust) that collects metrics via eBPF sched\_switch/sched\_process\_exec tracepoints instead of parsing /proc, cutting syscall cost from O(N) to O(1), measured via strace at ~780 constant syscalls per refresh vs htop's ~12,000 at 5,000 processes.
- Reduced hotpath cost per scheduler event from ~630ns to ~335ns across four A/B-validated optimizations; traced a 3x regression not to code but to the kernel silently downgrading the clocksource from TSC to HPET mid-benchmark.
- Hand-rolled CO-RE support by writing a BTF reader that resolves task\_struct field offsets at load time (aya doesn't emit CO-RE relocations); renders via a lock-free double-buffer (arc-swap) between the eBPF collector and the TUI.

**aya-rs (eBPF framework, Rust) - open-source contribution coursework, 19/20** | Mar 2026 – Present

- Fixed a silent ABI mismatch in aya-obj: expanded bpf\_map\_def from 28 to 36 bytes to match the bpf\_elf\_map layout used by the iproute2/tc toolchain, with full backward-compatible parsing ([PR #1497](#)).
- Implemented set\_tc\_classid on TcContext/SkBuff in aya-ebpf, enabling TC classifier programs to route packets to HTB/CBQ traffic-control classes, write-only by design, since the kernel verifier rejects reads of tc\_classid for sched\_cls program types ([PR #1595](#), co-authored with Afonso Quintas).

## EXPERIENCE

**RQ SYSTEMS - Co-Founder** | Jan 2026 – Present

- Architected an Outlook add-in (Office.js host, React/TypeScript frontend, n8n-orchestrated backend) that archives project emails and attachments to Dropbox, with Claude-based triage and structured data extraction for invoices and adjudications; deployed to Alphalink's full 30-person team.
- Designed the request flow to return AI-extracted data to the user immediately after analysis, deferring the archive step to run in the background, minimizing perceived latency.
- Specified a Dockerized microservice backend (n8n, Gotenberg, a Word-template generator) with no ports exposed outside the internal Docker network, and a versioned JSON contract (Zod-validated on both sides) between frontend and backend.

## TECHNICAL SKILLS

**Programming:** C, C++ (compilers coursework: lexers, parsers), Java, SQL, Rust, Bash.

**Systems & Infrastructure:** Linux (Arch/Debian administration), Docker, eBPF, Tailscale (VPN/Networking), Git.

**Tools & Frameworks:** n8n (Workflow Automation), PostgreSQL, Microsoft Graph API.

**Languages and certifications:** English (Certificate in Advanced English / C1 - Fluent), Portuguese (Native)

July 2026